

Yingkai Li

CONTACT INFORMATION	AS2 05-21 1 Arts Link, Singapore 117568	yk.li@nus.edu.sg https://yingkai-li.github.io Last updated: May 2026
RESEARCH INTERESTS	mechanism design, information design, microeconomic theory, algorithmic game theory	
EMPLOYMENT	Assistant Professor (Presidential Young Professor) in Economics, National University of Singapore	2024 -
	Postdoc Associates, Cowles Foundation for Research in Economics and Department of Computer Science, Yale University	2022 - 2024
	Research Intern, Microsoft Research	Jun to Aug 2020, 2021
EDUCATION	Northwestern University, Evanston, IL, USA	June 2022
	Ph.D., Computer Science Advisor: Jason D. Hartline	
	Stony Brook University, Stony Brook, NY, USA	May 2018
	M.S., Computer Science	
	Shanghai Jiaotong University, Shanghai, China	June 2015
	B.S., Computer Science	
AWARDS	Northwestern Terminal Year Fellowship	2021
WORKING PAPERS	1. Learning and Communication Towards Unanimous Consent. <i>with Boli Xu</i> revise and resubmit at American Economic Journal: Microeconomics 2. Selling Data to an Agent with Endogenous Information. major revision at Management Science 3. Managing Persuasion Robustly: The Optimality of Quota Rules. <i>with Dirk Bergemann and Tan Gan</i> revise and resubmit at Journal of Economic Theory 4. Mechanism Design under Costly Signaling: the Value of Non-Coordination. <i>with Xiaoyun Qiu</i> revise and resubmit at Theoretical Economics 5. Budget Pacing in Repeated Auctions: Regret and Efficiency without Convergence. <i>with Jason Gaitonde, Bar Light, Brendan Lucier and Alex Slivkins</i> major revision at Operations Research 6. Screening for Choice Sets. <i>with Tan Gan</i> 7. Anonymous Pricing in Large Markets. <i>with Yaonan Jin</i>	

8. Multi-Dimensional Screening with Endogenous Information Disclosure.
with Yang Cai and Jinzhao Wu
9. Incentivizing Forecasters to Learn: Summarized vs. Unrestricted Advice.
with Jonathan Libgober
10. Simple Mechanisms for Agents with Non-linear Utilities.
with Yiding Feng and Jason Hartline
11. Transparency in Team Contracts
with Paul Dütting, Inbal Talgam-Cohen and Jinzhao Wu
12. Mechanism Design with Endogenous Principal Learning.
with Daniel Clark
13. Fair Team Contracts.
with Matteo Castiglioni and Junjie Chen
14. Fair Contracts.
with Matteo Castiglioni and Junjie Chen
15. Scale-robust Auctions.
with Jason Hartline and Aleck Johnsen
16. Dynamics and Contracts for an Agent with Misspecified Beliefs.
with Argyris Oikonomou
17. Exploration and Incentivizing Participation in Randomized Trials.
with Alex Slivkins
18. Misspecified Beliefs about Time Lags.
with Harry Pei

JOURNAL
PUBLICATIONS

1. Information Elicitation Mechanisms for Bayesian Auctions.
with Jing Chen and Bo Li
Autonomous Agents and Multi-Agent Systems 2025
2. Nearly Minimax-Optimal Regret for Linearly Parameterized Bandits.
with Yining Wang and Yuan Zhou
Transactions on Information Theory 2024
3. Your College Dorm and Dormmates: Fair Resource Sharing with Externalities.
with Jiarui Gan and Bo Li
Journal of Artificial Intelligence Research 2023
4. Bayesian Auctions with Efficient Queries.
with Jing Chen, Bo Li and Pinyan Lu
Artificial Intelligence 2022
5. Equilibrium Behaviors in Repeated Games.
with Harry Pei
Journal of Economic Theory 2021
6. Efficient Approximations for the Online Dispersion Problem.
with Jing Chen and Bo Li
SIAM Journal on Computing 2019

1. Screening for Choice Sets.
with Tan Gan, EC 2026
2. Anonymous Pricing in Large Markets.
with Yaonan Jin, EC 2026
3. Learning and Communication Towards Unanimous Consent.
with Boli Xu, EC 2026
4. Allocating Resources under Strategic Misrepresentation
with Xiaoyun Qiu, EC 2026
5. Multi-Project Contracts.
with Tal Alon, Matteo Castiglioni, Junjie Chen, Tomer Ezra and Inbal Talgam-Cohen, EC 2025
6. Competition Complexity in Multi-item Auctions: Beyond VCG and Regularity.
with Hedyeh Beyhaghi, Linda Cai, Yiding Feng and Matthew Weinberg, EC 2025
7. Algorithmic Information Disclosure in Optimal Auctions.
with Yang Cai and Jinzhao Wu, EC 2024
8. Optimal Scoring for Dynamic Information Acquisition.
with Jonathan Libgober, EC 2024
9. Revenue Maximization for Buyers with Costly Participation.
with Yannai Gonczarowski, Nicole Immorlica and Brendan Lucier, SODA¹ 2024
10. Optimal Scoring Rules for Multi-dimensional Effort.
with Jason Hartline, Liren Shan and Yifan Wu, COLT² 2023
11. Bayesian Analysis of Linear Contracts.
with Tal Alon, Paul Dütting and Inbal Talgam-Cohen, EC³ 2023
12. Making Auctions Robust to Aftermarkets.
with Moshe Babaioff, Nicole Immorlica and Brendan Lucier, ITCS⁴ 2023
13. Budget Pacing in Repeated Auctions: Regret and Efficiency without Convergence.
with Jason Gaitonde, Bar Light, Brendan Lucier and Alex Slivkins, ITCS 2023
14. Simple Mechanisms for Non-linear Agents.
with Yiding Feng and Jason Hartline, SODA 2023
15. Selling Data to an Agent with Endogenous Information. EC 2022
16. Optimization of Scoring Rules.
with Jason Hartline, Liren Shan and Yifan Wu, EC 2022
17. Almost Proportional Allocations for Indivisible Chores.
with Bo Li and Xiaowei Wu, WebConf⁵ 2022

¹ACM-SIAM Symposium on Discrete Algorithms

²Conference on Learning Theory

³ACM Conference on Economics and Computation

⁴Innovations in Theoretical Computer Science

⁵The Web Conference

18. Revelation Gap for Pricing from Samples.
with Yiding Feng and Jason Hartline, STOC⁶ 2021
19. Tight Regret Bounds for Infinite-armed Linear Contextual Bandits.
with Yining Wang, Xi Chen and Yuan Zhou, AISTATS⁷ 2021
20. Benchmark Design and Prior-independent Optimization.
with Jason Hartline and Aleck Johnsen, FOCS⁸ 2020
21. Multinomial Logit Bandit with Low Switching Cost.
with Kefan Dong, Qin Zhang and Yuan Zhou, ICML⁹ 2020
22. Fair Resource Sharing and Dorm Assignments.
with Bo Li, AAMAS¹⁰ 2020
23. Approximately Maximizing the Broker's Profit in a Two-sided Market.
with Jing Chen and Bo Li, IJCAI¹¹ 2019
24. Optimal Auctions vs. Anonymous Pricing: Beyond Linear Utility.
with Yiding Feng and Jason Hartline, EC 2019
25. Nearly Minimax-Optimal Regret for Linearly Parameterized Bandits.
with Yining Wang and Yuan Zhou, COLT 2019
26. Revenue Maximization with Imprecise Distribution.
with Pinyan Lu and Haoran Ye, AAMAS 2019
27. Information Elicitation for Bayesian Auctions.
with Jing Chen and Bo Li, SAGT¹² 2018
28. Dynamic Fair Division Problem with General Valuations.
with Bo Li and Wenyang Li, IJCAI 2018
29. Bayesian Auctions with Efficient Queries (Brief Announcement).
with Jing Chen, Bo Li and Pinyan Lu, ICALP¹³ 2018
30. Efficient Approximations for the Online Dispersion Problem.
with Jing Chen and Bo Li, ICALP 2017

⁶ACM Symposium on Theory of Computing

⁷International Conference on Artificial Intelligence and Statistics

⁸IEEE Symposium on Foundations of Computer Science

⁹International Conference on Machine Learning

¹⁰International Conference on Autonomous Agents and Multiagent Systems

¹¹International Joint Conferences on Artificial Intelligence

¹²International Symposium on Algorithmic Game Theory

¹³EATCS International Colloquium on Automata, Languages and Programming

ACADEMIC SERVICE	Workflow Chair: EC 2024	
	Senior Program Committee	
	WINE¹⁴ 2025	
	Program Committee	
	EC 2025, WINE 2024, EAAMO 2024¹⁵, EC 2024, WINE 2023, EC 2023, WebConf 2023, WINE 2022	
	Journal Reviewer	
	American Economic Review, Econometrica, Review of Economic Studies, Journal of the ACM, American Economic Review: Insight, American Economic Journal: Microeconomics, Journal of Economic Theory, SIAM Journal on Computing, Operations Research, Mathematics of Operations Research, Games and Economic Behavior, Transactions on Information Theory, Transactions on Economics and Computation, Economic Theory, Artificial Intelligence, Theory and Decision.	
	Conference Reviewer	
	STOC, SODA, EC, ICALP, ICML, ITCS, KDD, AISTATS, ESA, WebConf, WINE, COCOA	
CURRENT STUDENTS	- Zeyu Wang (PhD) - Junjie Chen (Postdoc) - Yiyao Zhu (Postdoc)	
FORMER STUDENTS	- Naman Agrawal (Undergrad), then Predoc at NUS. - Hanyang Jiao (Undergrad), then Master at NUS.	
TEACHING EXPERIENCE	Instructor - National University of Singapore	
	EC4501/EC4501HM Economics and Computation	Spring 2025
	EC5881/EC5881R - Topics in Microeconomics	Fall 2024
	Teaching Assistant - Northwestern University	
	COMP_SCI 396 - Online Markets	Spring 2020
	COMP_SCI 336 - Design & Analysis of Algorithms	Fall 2019
	COMP_SCI 212 - Mathematical Foundations of Computer Science	Spring 2019
	Teaching Assistant - Stony Brook University	
	CSE 215 - Foundations of Computer Science	Fall 2015, Spring 2016
	CSE 114 - Computer Science I	Spring 2016
	CSE 540 - Theory of Computation	Fall 2016

¹⁴Conference on Web and Internet Economics

¹⁵ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization